

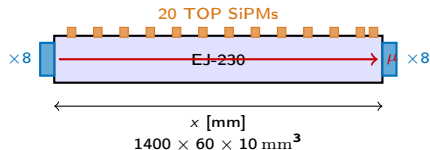
SHiP Timing Bar:
END+TOP Hybrid T_0 Estimator
EJ-230 Scintillator, Vikuiti Reflector ($R = 0.95$)

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- **END**: 2×8 SiPMs on bar faces $\Rightarrow$ 16 channels
- **TOP**: N_{TOP} SiPMs on top surface
- Vikuiti ESR reflector on all lateral faces (dielectric-dielectric, $R = 0.95$ constant)

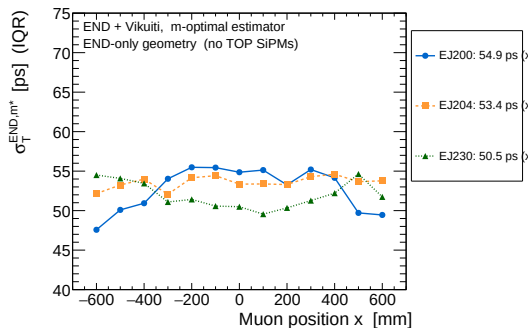
Geant4 simulation

- Muons shot at 13 positions
 $x \in [-600, +600]$ mm
- 5000 events per position
- Branches: event_id, face_type, time_ns, pde, x_mm, gun_x_mm
- face_type: 0=END-L, 1=END-R, 2=TOP
- SiPM: wavelength-dependent PDE

Material: EJ-230

Attenuation	1200 mm
Decay time	1.5 ns
Rise time	0.5 ns
Yield	9700 MeV^{-1}
n	1.58

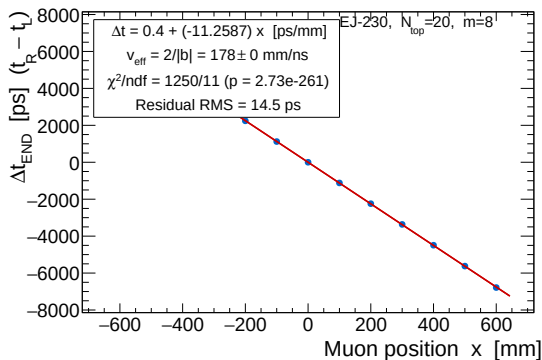
EJ-230 chosen for its fast decay — material comparison



Scintillator properties

	EJ-200	EJ-204	EJ-230
Att. [mm]	3800	1600	1200
τ [ns]	2.1	1.8	1.5
t_r [ns]	0.9	0.7	0.5
Y [/MeV]	10000	10400	9700
n	1.58	1.58	1.58

- EJ-230: fastest rise ($t_r = 0.5$ ns) and decay ($\tau = 1.5$ ns)
- Shortest attenuation — photon loss toward center, but timing wins
- END-only at $x = 0$: EJ-230 best despite fewer photons
- All three use $n = 1.58$, $R = 0.95$ reflector



Linear fit: $\Delta t = a + bx$

$$b = -11.259 \pm 0.001 \text{ ps/mm}$$

$$v_{\text{eff}} = \frac{2}{|b|} = 177.6 \pm 0.0 \text{ mm/ns}$$

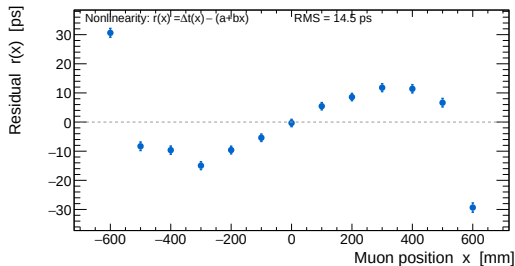
$$\chi^2/\text{ndf} = 1250/11$$

Significant nonlinearity

$\chi^2/\text{ndf} = 1250/11$ is *not* mild — $p \approx 0$.
 The linear fit is a first-order approximation;
 residuals shown next.

- Used: $m = 8$ average estimator for Δt
- $\Delta t = t_R^{(m)} - t_L^{(m)}$
- Symmetry: $a \approx 0$ confirms bar is centred

Nonlinearity residual has S-shape — edge photon loss

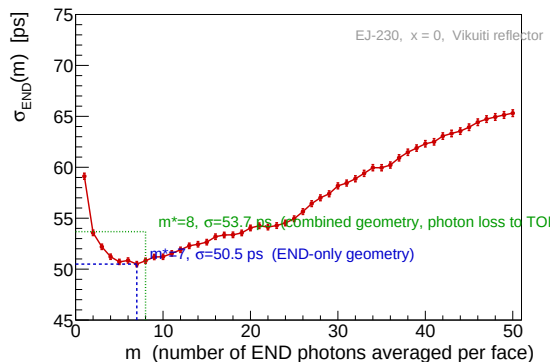


$$\text{Residual } r(x) = \langle \Delta t \rangle - (a + bx)$$

RMS = 14.5 ps

x [mm]	r [ps]
-600	+30.6
-300	-15.0
0	-0.3
+300	+11.8
+600	-29.4

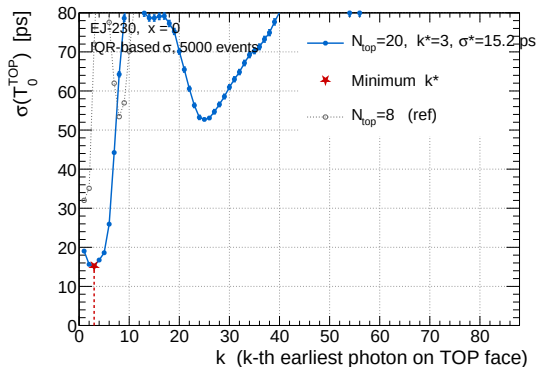
- S-shape: near $x = \pm 600$ mm photons arrive at the *near* end first, biasing the average
- Residual RMS ≈ 15 ps is the nonlinearity contribution to position bias
- For position reconstruction, a cubic correction should be applied



m -average estimator

$$T_0^{\text{END}} = \frac{1}{2}(t_L^{(m)} + t_R^{(m)}), \quad t_\alpha^{(m)} = \frac{1}{m} \sum_{i=1}^m t_{\alpha,i}$$

- Sort hits per event per face; take first m arrival times
- Averaging over m photons suppresses single-photon jitter
- Global optimum: $m = 8$ (train/test at $x = 0$)
- $\sigma_{\text{END}}(x = 0, m = 8) = 53.7$ ps
- σ_{END} rises at bar ends as near-face photon pool shrinks



k -th order statistic (pooled)

$$T_0^{\text{TOP}} = t_{(k)}, \quad t_{(1)} \leq t_{(2)} \leq \dots$$

Pool *all* N_{TOP} SiPMs; pick k -th earliest.

- Global optimum: $k = 3$ (train/test at $x = 0$)
- $\sigma_{\text{TOP}}(x = 0, k = 3) = 15.2 \text{ ps}$
- $3.6\times$ improvement over per-SiPM mean
- Low k exploits first-photon statistics without dark-count sensitivity
- Insensitive to x : same photons reach TOP from all muon positions

BLUE combination

Given estimators T_E , T_T with covariance matrix

$$C = \begin{pmatrix} \text{Var}(T_E) & \text{Cov}(T_E, T_T) \\ \text{Cov}(T_E, T_T) & \text{Var}(T_T) \end{pmatrix}$$

BLUE weight (ordinary Var/Cov):

$$w_E = \frac{\text{Var}(T_T) - \text{Cov}}{\text{Var}(T_E) + \text{Var}(T_T) - 2 \text{Cov}}, \quad w_T = 1 - w_E$$

$$T_0^{\text{BLUE}} = w_E T_E + w_T T_T$$

$\sigma_{\text{IQR}} = (Q_{75} - Q_{25})/1.349$ is a **separate** robust performance metric — not derived from C .

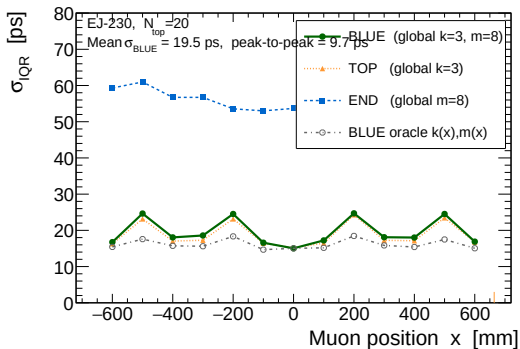
At $x = 0$, $k = 3$, $m = 8$

Quantity	Value
σ_{END}	53.7 ps
σ_{TOP}	15.2 ps
σ_{BLUE}	15.0 ps
w_{END}	0.086
w_{TOP}	0.914
$\rho(E, T)$	0.160

Interpretation

$w_{\text{END}} \approx 0.09$ — when TOP has $\sigma_{\text{TOP}} \approx 15$ ps, END contributes only 9% of the weight. BLUE $\approx$ TOP at this configuration.

Global estimator is deployable; oracle is a lower bound



Two estimator classes

Global ($k=3$, $m=8$ fixed $\forall x$):
Deployable without per-position tuning

Oracle (local opt. $k^*(x)$, $m^*(x)$):
Diagnostic lower bound — not deployable

Results across $x \in [-600, +600]$ mm

	Mean	Min	Max
Global σ_{BLUE}	19.5	15.0	24.7
Oracle σ_{BLUE}	16.1	14.7	18.5

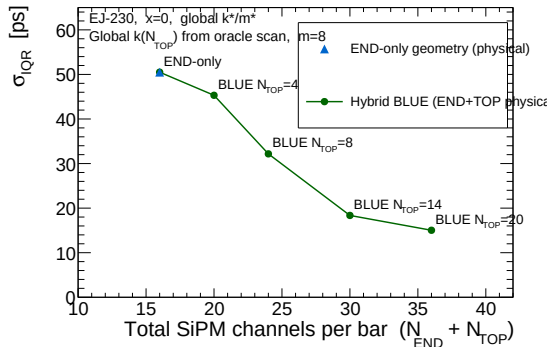
Units: ps

- Spikes at $x = \pm 200, \pm 500$ mm where $k^*(x) = 1$ but global uses $k = 3$
- Oracle gap: ~ 3.4 ps (mean)
- Tick marks = TOP SiPM positions

x	σ_{END}	σ_{TOP}	σ_{BLUE}	note
-600	59.3	16.3	16.7	edge
-500	61.0	23.2	24.6	spike
-400	56.7	17.0	18.1	
-300	56.7	17.3	18.6	
-200	53.6	23.2	24.5	spike
-100	53.0	16.4	16.6	
0	53.7	15.2	15.0	
+100	53.8	16.5	17.2	
+200	55.0	24.3	24.7	spike
+300	55.3	17.2	18.1	
+400	57.6	17.1	18.0	
+500	59.7	23.5	24.5	spike
+600	61.0	16.8	16.9	edge

Units: ps; global $k = 3$, $m = 8$

- Spikes at $x = \pm 200, \pm 500$ mm: these lie between TOP SiPMs, so the $k = 3$ photon arrives later than when x is near a SiPM
- Oracle uses $k^*(x) = 1$ at these positions, recovering $\sigma_{\text{TOP}} \approx 17\text{-}18$ ps
- No spike in END timing — it responds monotonically with distance
- Reducing k globally (e.g. $k = 1$) would help at spikes but hurt at SiPM positions
- Trade-off is fundamental to fixed-parameter deployment

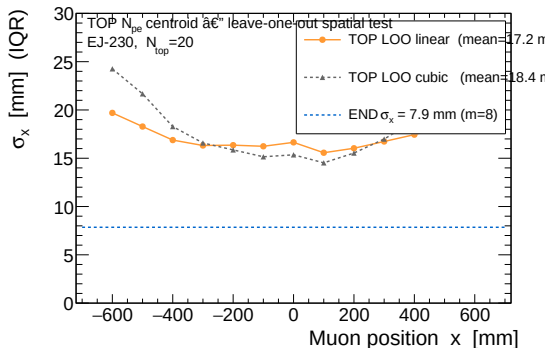


Physical geometries, $x = 0$

N_{TOP}	k^*	σ_{TOP}	σ_{BLUE}	w_E
4	1	78.2	45.3	0.77
8	1	32.0	32.2	0.44
14	2	18.7	18.4	0.17
20	3	15.2	15.0	0.09

ps; $m = 8$ fixed; k^* from oracle scan

- $N_{TOP} = 4$: END dominates ($w_E = 0.77$)
- $N_{TOP} = 14$: $\sigma_{BLUE} \approx 18$ ps, $3\times$ fewer channels than $N_{TOP} = 20$
- $N_{TOP} = 20$: $\sigma_{BLUE} = 15.0$ ps, END weight collapses to 9%
- k^* grows with N_{TOP} : more SiPMs means more photons to pick from



N_{pe} -weighted centroid

$$x_c = \frac{\sum_i N_{pe,i} x_i^{SiPM}}{\sum_i N_{pe,i}}$$

- SiPMs span ± 664 mm; centroid compressed to ± 313 mm at bar edges
- LOO: train on 12 positions, predict 13th
- **Linear**: mean $\sigma_x = 17.2$ mm, max $|\text{bias}| = 29$ mm at ends
- **END** Δt position: $\sigma_x = 7.9$ mm (from v_{eff})
- END is $2\times$ better and has no edge bias
- TOP centroid still useful for tracking in a multi-bar system

EJ-230 + END+TOP hybrid

- EJ-230: fastest scintillator ($\tau = 1.5$ ns, att= 1200 mm)
- Reflector: $R = 0.95$ constant (Vikuiti ESR, dielectric-dielectric)
- $v_{\text{eff}} = 177.6$ mm/ns, $\chi^2/\text{ndf} = 1250/11$ — statistically significant nonlinearity (RMS= 14.5 ps)

Timing performance

- END ($m = 8$): $\sigma_{\text{END}} = 53.7$ ps at $x = 0$
- TOP ($k = 3$, $N_{\text{TOP}} = 20$): $\sigma_{\text{TOP}} = 15.2$ ps
- BLUE global: $\sigma_{\text{BLUE}} = 15.0$ ps at $x = 0$, mean = 19.5 ps across full bar
- Oracle: mean 16.1 ps — 3.4 ps gap shows cost of fixed parameters

N_{TOP} trade-off

- $N_{\text{TOP}} = 4$: 45.3 ps (END-dominated)
- $N_{\text{TOP}} = 14$: 18.4 ps ($3\times$ fewer ch.)
- $N_{\text{TOP}} = 20$: 15.0 ps (recommended)

Position

- END Δt : $\sigma_x = 7.9$ mm
- TOP centroid (LOO): $\sigma_x = 17.2$ mm, large bias at edges — END preferred

Next steps

- Cubic correction for Δt nonlinearity
- Systematic: dark counts, SiPM cross-talk
- Test on beam data

Recorded PDE in simulation

- The SiPM model applies a wavelength-dependent efficiency curve
- Each *detected* photon carries its efficiency as the pde branch
- Mean over **detected photons**: $\langle \text{PDE} \rangle = 0.607$

Selection bias

This is *not* the detection fraction. High-PDE photons are over-represented because they are more likely to be accepted. The correct photon budget uses $N_{\text{detected}}/N_{\text{emitted}}$.

By face ($x=0$, EJ-230, $N_{\text{TOP}} = 20$)

Face	$\langle \text{PDE} \rangle_{\text{det}}$	N_{pe}/ev
END-L	0.608	515
END-R	0.608	515
TOP	0.606	1341
All	0.607	—

- 11.85M detected hits at $x = 0$
- TOP captures $\sim 2.6\times$ more photons than each END face
- More photons support lower k on TOP

Geant4 surface: CreateBarSkinReflector()

- Type: `dielectric_dielectric`
 - Model: `unified`
 - Finish: `polished`
 - Reflectivity: $R = 0.95$ (constant, wavelength-independent)
-
- Source: `src/Materials.cc, CreateBarSkinReflector()`
 - Lateral faces wrapped in air-gap + Mylar in the physical geometry
 - *Not* the full Vikuiti ESR spectral table — a conservative constant

SiPM surface: wavelength-dependent PDE

- `SiPMModel::LoadPDECurve()`
- Applied at each photon absorption step
- Results in $\langle \text{PDE} \rangle = 0.607$ over detected population (selection-biased)

Implication

The reflector is a constant, not the Vikuiti spectral curve. Real Vikuiti $R \sim 0.98\text{--}0.99$ would yield more photons and better timing.

Oracle $k^*(x)$ and $m^*(x)$ vary across positions

x	k^*	σ_{TOP}^*	m^*	σ_{END}^*	σ_{BLUE}^*
-600	2	15.4	10	59.0	15.4
-500	1	17.8	11	60.1	17.6
-400	2	15.9	5	55.4	15.7
-300	2	16.0	6	56.3	15.6
-200	2	17.6	8	53.6	18.3
-100	2	14.9	6	52.8	14.7
0	3	15.2	8	53.7	15.0
+100	2	15.6	5	53.1	15.2
+200	2	18.2	10	54.5	18.5
+300	2	15.8	7	54.9	15.8
+400	2	15.8	5	57.4	15.4
+500	1	17.5	5	58.8	17.5
+600	2	15.3	5	60.9	15.1

ps; oracle = same-sample optimum

- Oracle $k^* = 1$ at $x = \pm 500$ mm (interstitial, few early photons in TOP)
- Oracle $k^* = 3$ only at $x = 0$ where global = oracle
- Global $k = 3$ is not optimal at most positions — hence the spikes in global σ_{BLUE}
- Oracle m^* ranges 5–11: END optimal m also varies with photon count
- Oracle diagnostic confirms the fundamental limit $\sigma_{\text{BLUE}} \approx 15$ ps is achievable at all positions (with position-dependent tuning)

50/50 split, $x = 0$, EJ-230, $N_{\text{TOP}} = 20$

Estimator	Train	Test
m^*	7	—
k^*	3	—
σ_{END} [ps]	53.7	54.0
σ_{TOP} [ps]	15.6	15.0
σ_{BLUE} [ps]	(cross)	14.8
w_{END}	—	0.020
ρ	—	0.141

- Optimal $m^* = 7$ (train); global uses $m = 8$
- At $x = 0$: global and oracle agree — $k = 3$, $m = 8$ is near-optimal

Gaussian fit at $x = 0$ (oracle $k = 3$, $m = 8$)

Est.	σ_{IQR}	σ_G	χ^2/ndf
END	53.7	53.3	65.8/52
TOP	15.2	14.8	62.0/53
BLUE	15.2	14.6	83.6/53

ps; BLUE non-Gaussian ($p = 0.005$)

- END and TOP are consistent with Gaussian
- BLUE slightly non-Gaussian ($p = 0.005$) due to the nonlinear weight combination
- σ_{IQR} is the robust metric; Gaussian σ shown for comparison only

- TOP SiPMs span $x \in [-664, +664]$ mm (spacing ≈ 70 mm)
- Centroid is bounded by the SiPM range
- At $x = \pm 600$ mm muon, median centroid is only ± 313 mm — compression by $2\times$
- Linear calibration has maximum $|\text{bias}| = 29$ mm at bar edges
- Cubic calibration reduces bias but does not improve σ_x
- Both are worse than END Δt ($\sigma_x = 7.9$ mm) at all positions

LOO centroid medians

x_{true}	$\bar{x}_c$	σ_c [mm]
-600	-313	10.7
-500	-273	9.9
-400	-218	9.1
-300	-168	8.8
-200	-121	8.8
-100	-58	8.8
0	+0	9.0
+100	+58	8.4
+200	+120	8.6
+300	+168	9.0
+400	+218	9.4
+500	+273	9.9
+600	+313	10.5